

Volcano Quest Flutter Development Roadmap

Target Deadline: July 2026

Platform: Android (Offline-first educational game)

Main Engine: Flutter

This roadmap is optimized for a solo developer building an educational gamified learning application with lightweight 3D integration.

Recommended Architecture

Main Frontend: Flutter

State Management: Riverpod

Routing: go_router

Offline Storage: Hive or SharedPreferences

3D Integration: model_viewer_plus

Animations: Lottie / Rive

Audio: just_audio

Firebase vs Supabase Recommendation

Since your application is primarily offline and cloud sync is only needed for dashboards, Firebase is the better option for your project.

Why Firebase fits better:

- Easier offline synchronization
- Better Flutter ecosystem support
- Faster setup for solo developers
- Realtime Database and Firestore work well for leaderboard syncing
- Excellent documentation for Flutter
- Stable Android support

Why Supabase may not be ideal here:

- More backend-oriented setup
- Better for SQL-heavy systems and web apps
- You may spend more time managing backend architecture

Recommended Setup:

- Offline gameplay stored locally using Hive
- Firebase only for dashboard, leaderboard, and analytics sync

Recommended Flutter Packages

- flutter_riverpod
- go_router
- hive
- hive_flutter
- firebase_core
- cloud_firestore
- firebase_analytics
- model_viewer_plus
- lottie
- rive
- just_audio
- flutter_animate
- confetti

Phase 1 — Planning & Setup (May 18 – May 24)

Tasks:

- Setup Flutter project
- Create GitHub repository
- Design UI in Figma
- Setup Firebase project
- Plan navigation flow
- Create XP and badge system structure
- Create folder architecture

Phase 2 — Core System Development (May 25 – June 7)

Tasks:

- Build offline storage system
- Build XP system
- Build badge system
- Build progress tracking
- Build audio manager
- Build reusable question engine
- Build leaderboard sync

Phase 3 — Game Level Development (June 8 – June 25)

Level 1: Research Base

- Tap interactions
- Animated glowing objects

Level 2: Knowledge Scanner

- Multiple choice system
- Score validation

Level 3: Lava Word Builder

- Drag-and-drop letters
- Word detection

Level 4: Volcano Explorer

- Interactive volcano map
- Quiz popups

Level 5: Volcano Anatomy Lab

- 3D volcano model
- Draggable labels
- Label detection system
- Magma animation effects

Level 6–8:

- Matching systems
- Quiz mechanics
- Warning sign activities

Phase 4 — Polish & Optimization (June 26 – July 5)

Tasks:

- Add sound effects
- Add achievement animations
- Improve responsiveness
- Optimize APK size
- Improve loading performance
- Improve low-end Android compatibility

Phase 5 — Testing & Bug Fixing (July 6 – July 15)

Tasks:

- Test progression flow
- Test all quiz systems
- Test drag-and-drop systems
- Fix crashes and bugs
- Test offline storage reliability
- Test Firebase dashboard synchronization

Phase 6 — Finalization (July 16 – July 20)

Tasks:

- Generate release APK
- Prepare screenshots
- Prepare presentation demo
- Prepare thesis/capstone documentation
- Final optimization pass

Recommended Asset Workflow

UI Design: Figma
3D Assets: Blender + AI-generated models
Animations: LottieFiles / Rive
Audio: Free educational sound assets

Recommended asset sources:

- Sketchfab
- Poly Pizza
- CGTrader
- Meshy AI

Important Development Advice

- Keep the 3D system lightweight
- Avoid advanced physics systems
- Focus on gameplay stability first
- Prioritize educational interaction quality over graphics
- Test regularly on low-end Android devices
- Finish MVP features before polishing visuals

Weekly Timeline Overview

Week	Focus
Week 1	Project setup and UI planning
Week 2	Core systems and offline architecture
Week 3	Educational levels implementation
Week 4	3D volcano anatomy integration
Week 5	Polish and optimization
Week 6	Testing and bug fixing
Week 7	Finalization and presentation prep